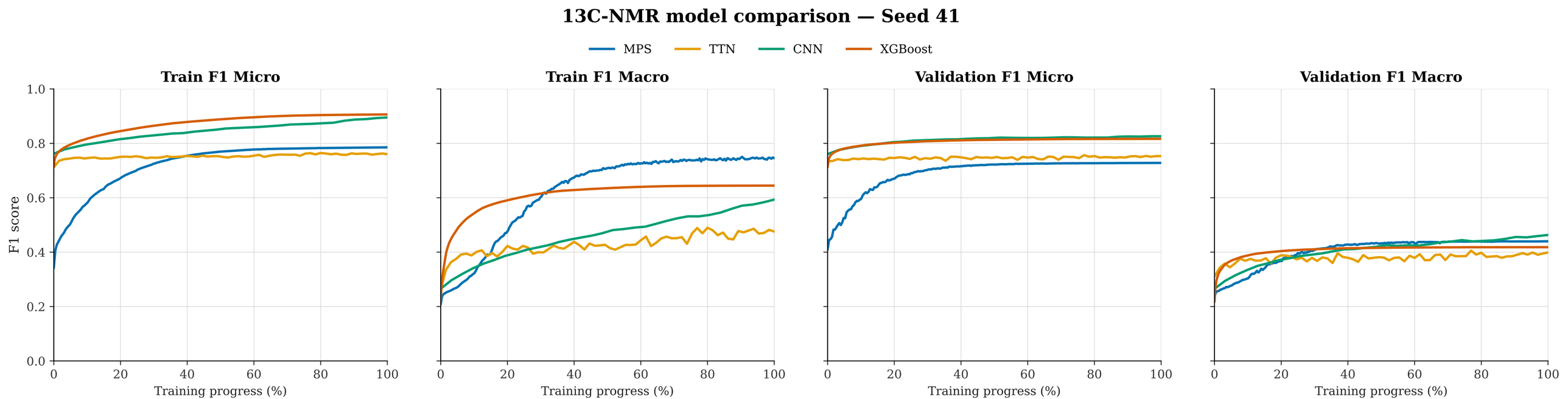
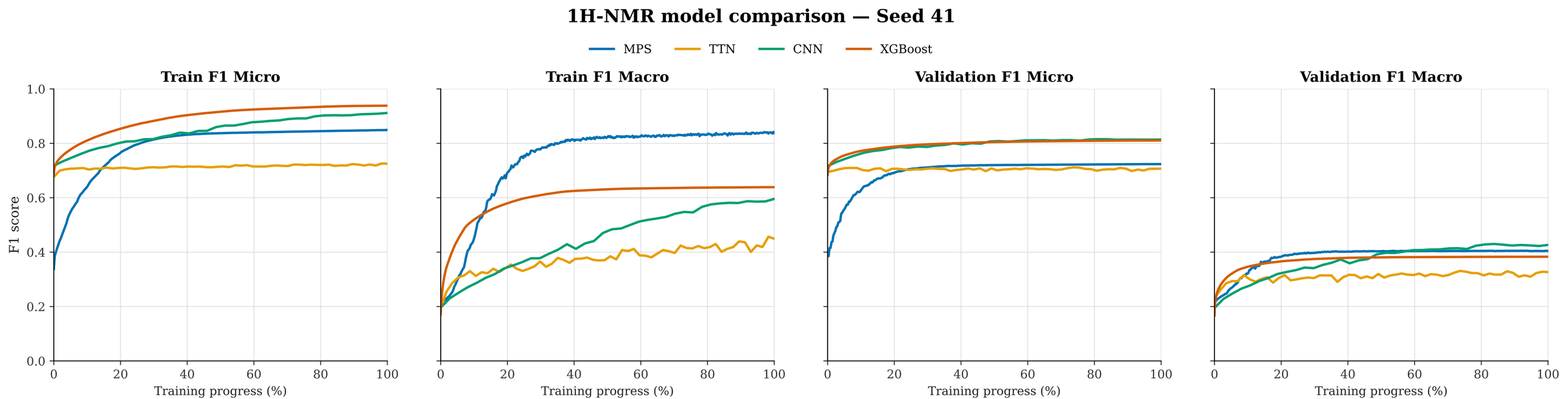
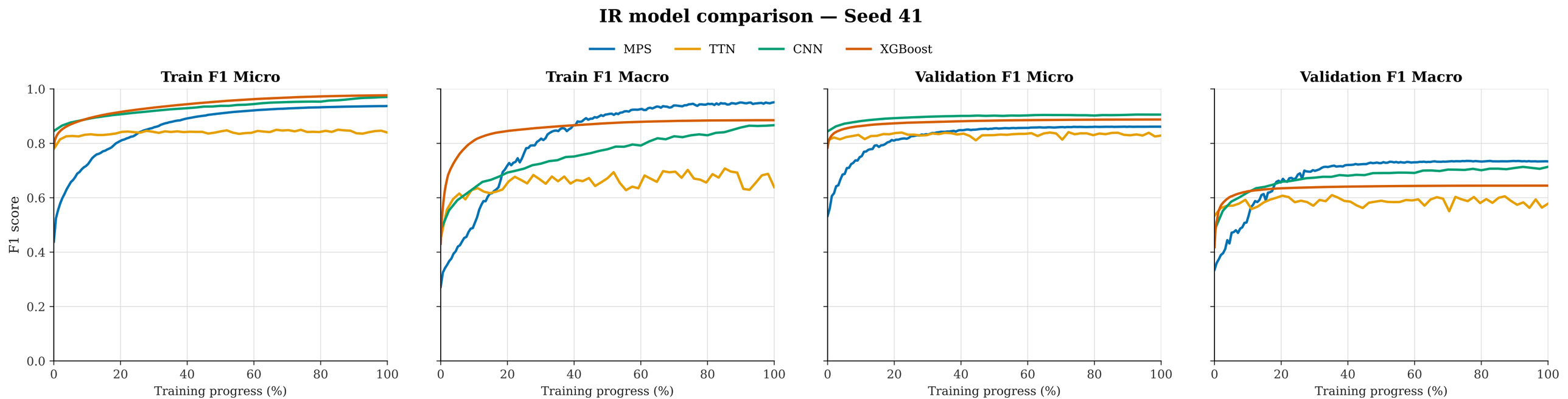
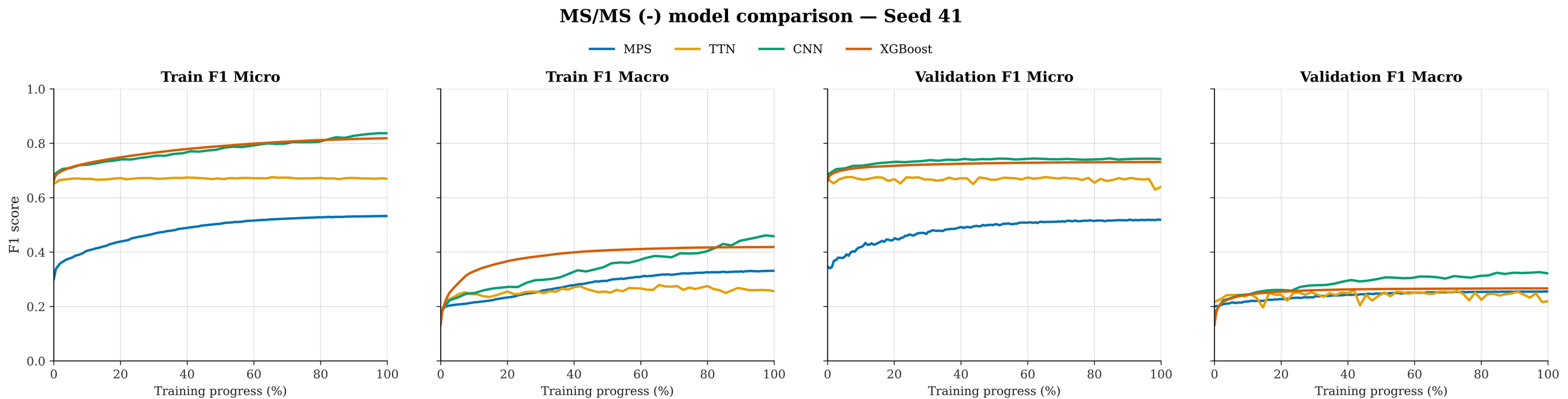
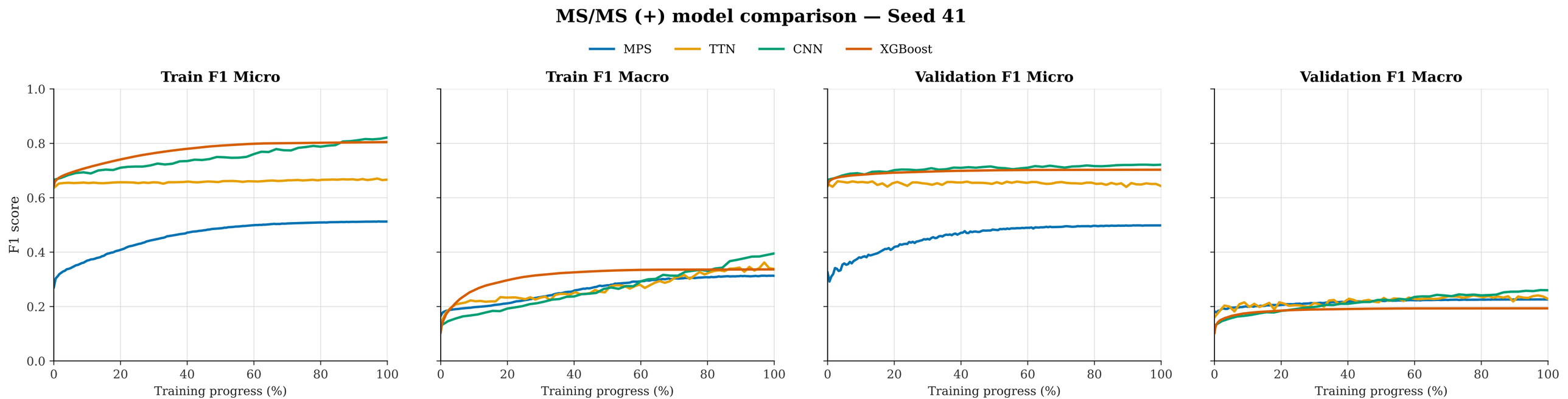
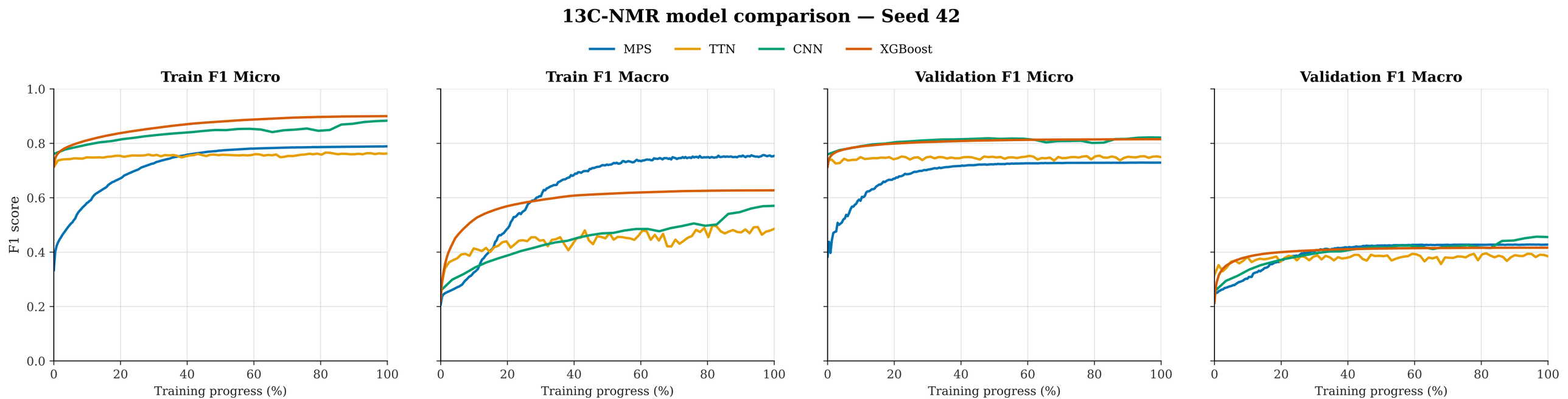

13C-NMR model comparison — Seed 41
MPS
TTN
CNN
XGBoost
Train F1 Micro
Train F1 Macro
Validation F1 Micro
Validation F1 Macro
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Training progress (%)
FIR + Gaussian smoothing + sqrt transformation + SNV normalization | Input dimension: 600

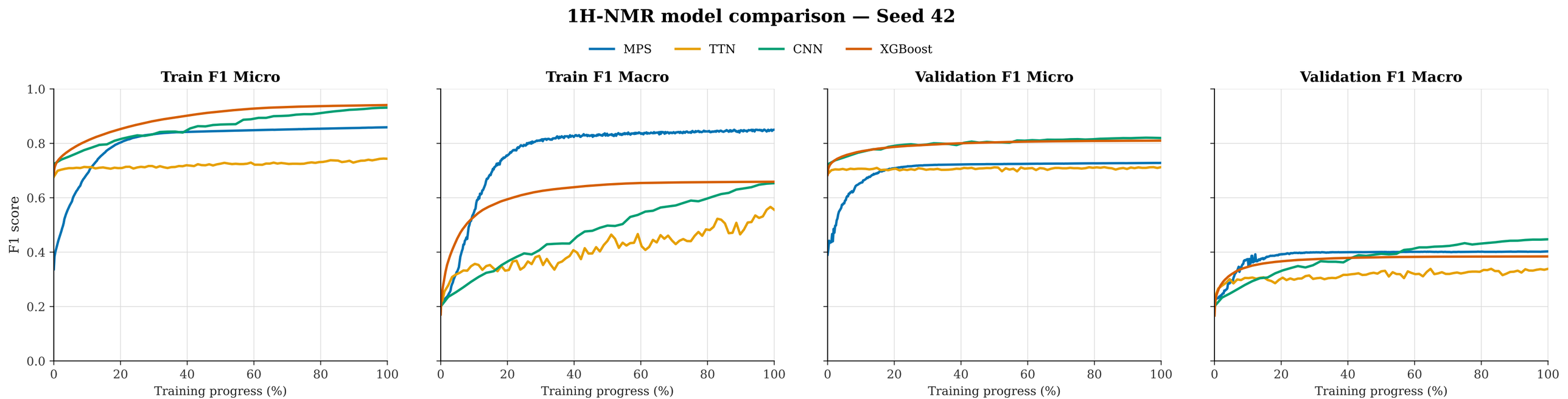

1H-NMR model comparison — Seed 41
MPS
TTN
CNN
XGBoost
Train F1 Micro
Train F1 Macro
Validation F1 Micro
Validation F1 Macro
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Training progress (%)
FIR + Gaussian smoothing + sqrt transformation + SNV normalization | Input dimension: 600

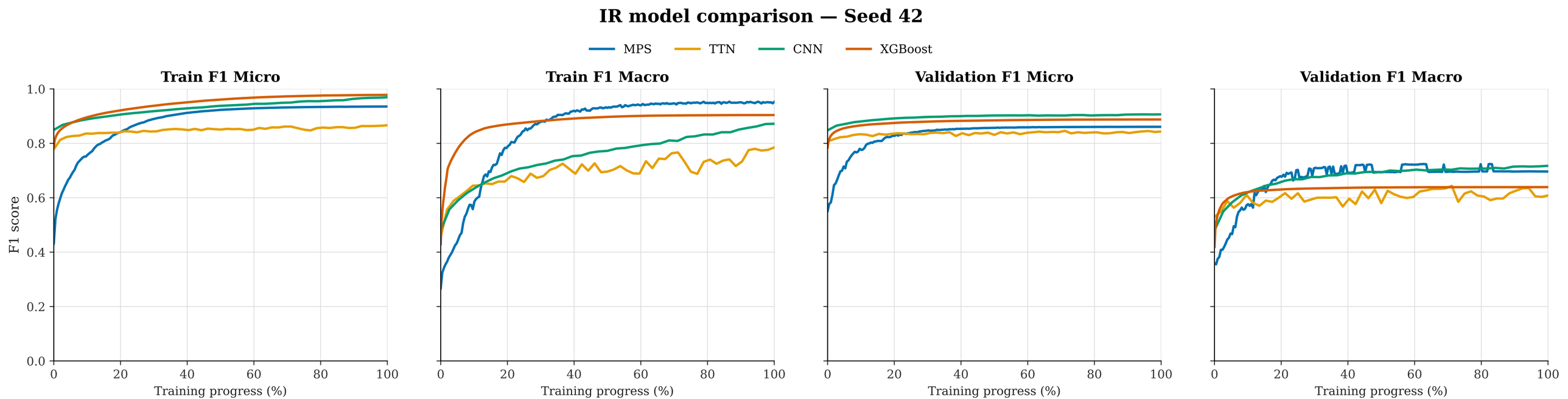

IR model comparison — Seed 41
MPS
TTN
CNN
XGBoost
Train F1 Micro
Train F1 Macro
Validation F1 Micro
Validation F1 Macro
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Training progress (%)
FIR + Gaussian smoothing + sqrt transformation + SNV normalization | Input dimension: 600

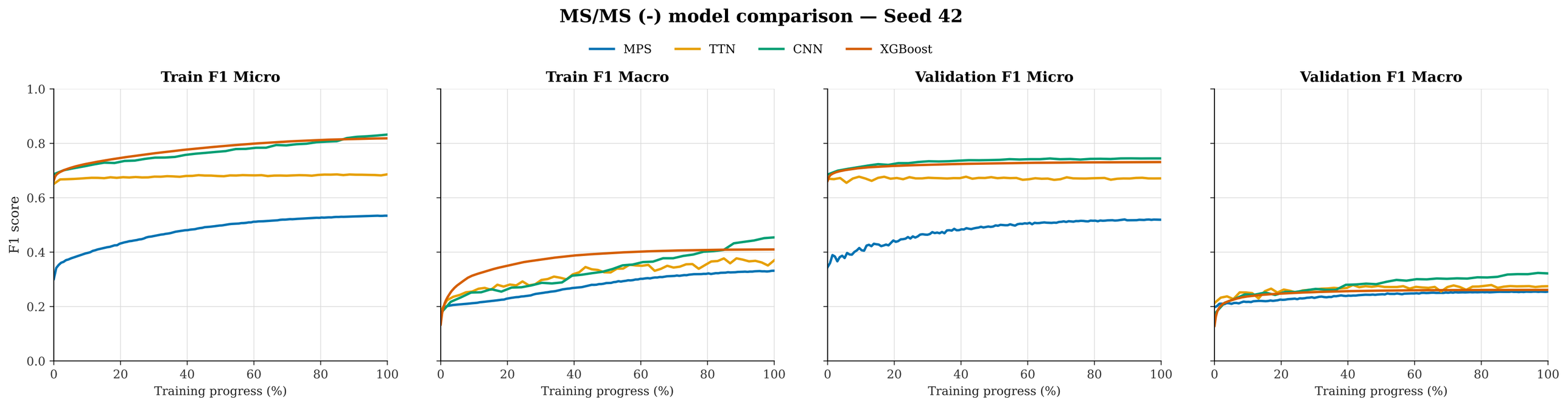

MS/MS (-) model comparison — Seed 41
MPS
TTN
CNN
XGBoost
Train F1 Micro
Train F1 Macro
Validation F1 Micro
Validation F1 Macro
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Training progress (%)
FIR + Gaussian smoothing + sqrt transformation + SNV normalization | Input dimension: 600

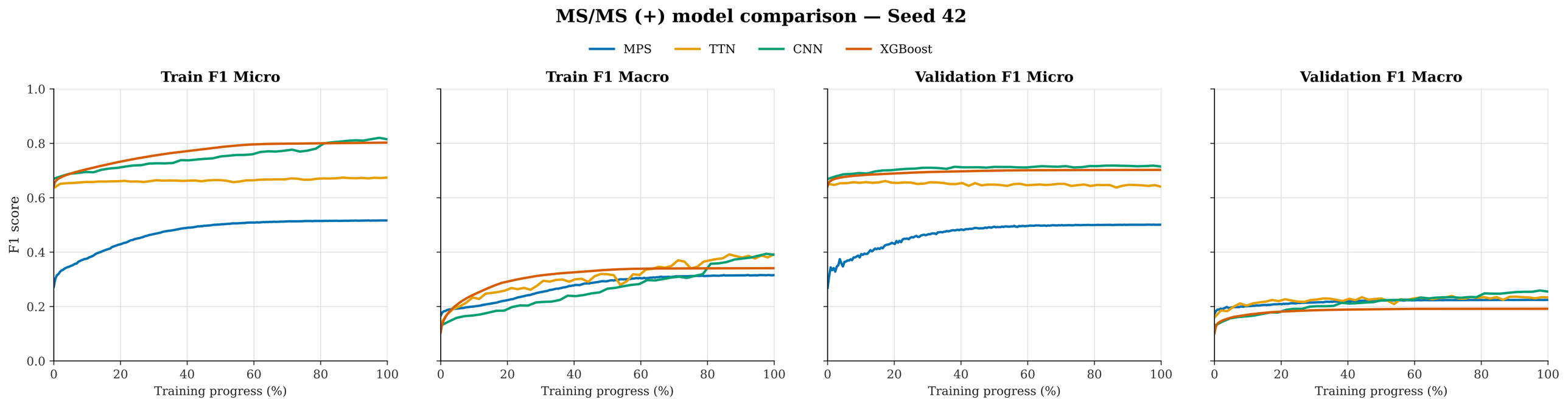

MS/MS (+) model comparison — Seed 41
MPS
TTN
CNN
XGBoost
Train F1 Micro
Train F1 Macro
Validation F1 Micro
Validation F1 Macro
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Training progress (%)
FIR + Gaussian smoothing + sqrt transformation + SNV normalization | Input dimension: 600

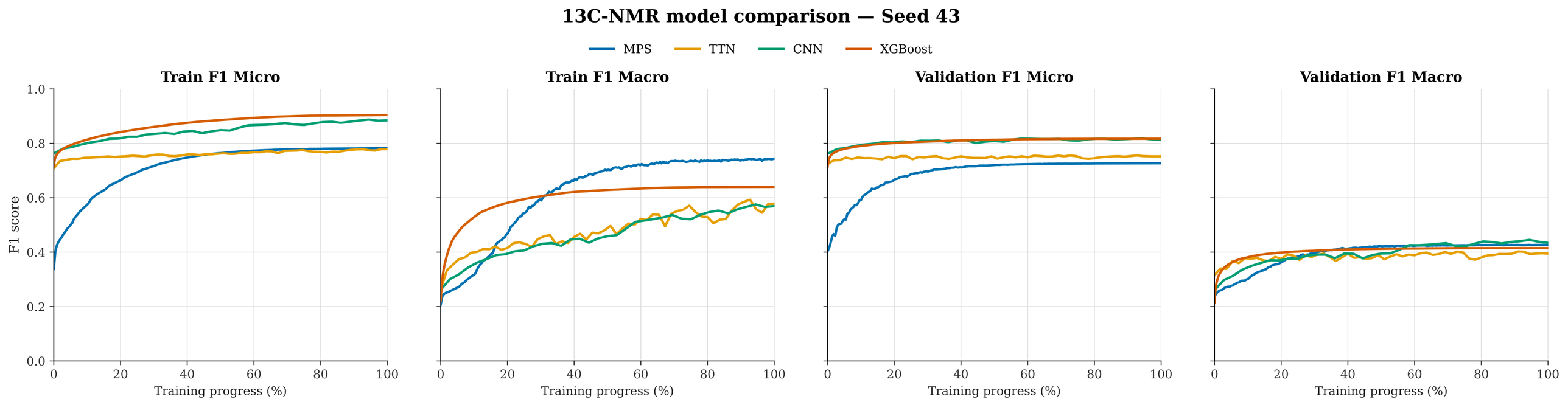

13C-NMR model comparison — Seed 42
MPS
TTN
CNN
XGBoost
Train F1 Micro
Train F1 Macro
Validation F1 Micro
Validation F1 Macro
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Training progress (%)
FIR + Gaussian smoothing + sqrt transformation + SNV normalization | Input dimension: 600

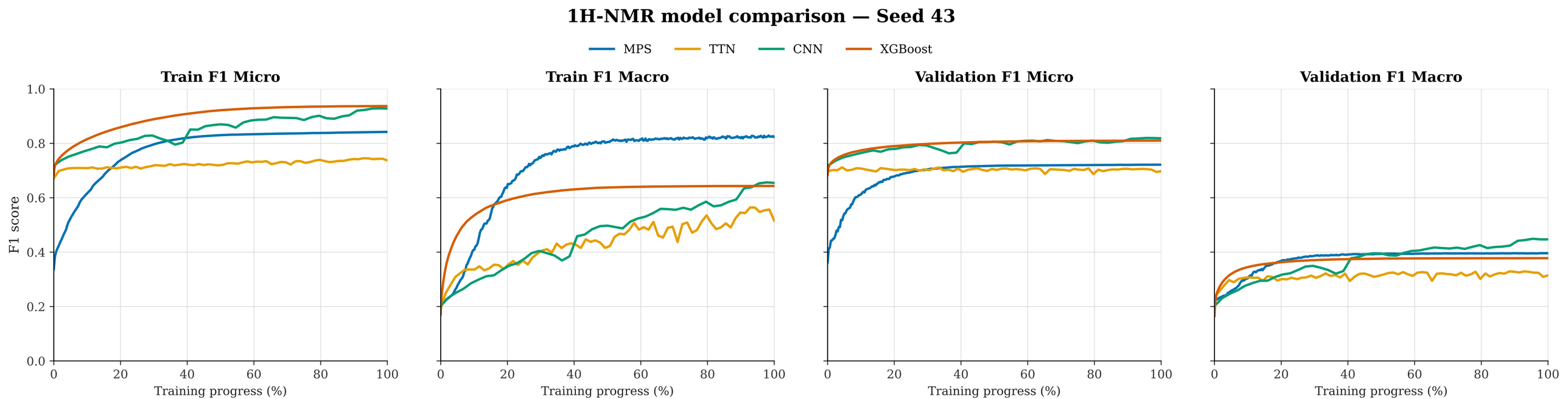

1H-NMR model comparison — Seed 42
MPS
TTN
CNN
XGBoost
Train F1 Micro
Train F1 Macro
Validation F1 Micro
Validation F1 Macro
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Training progress (%)
FIR + Gaussian smoothing + sqrt transformation + SNV normalization | Input dimension: 600

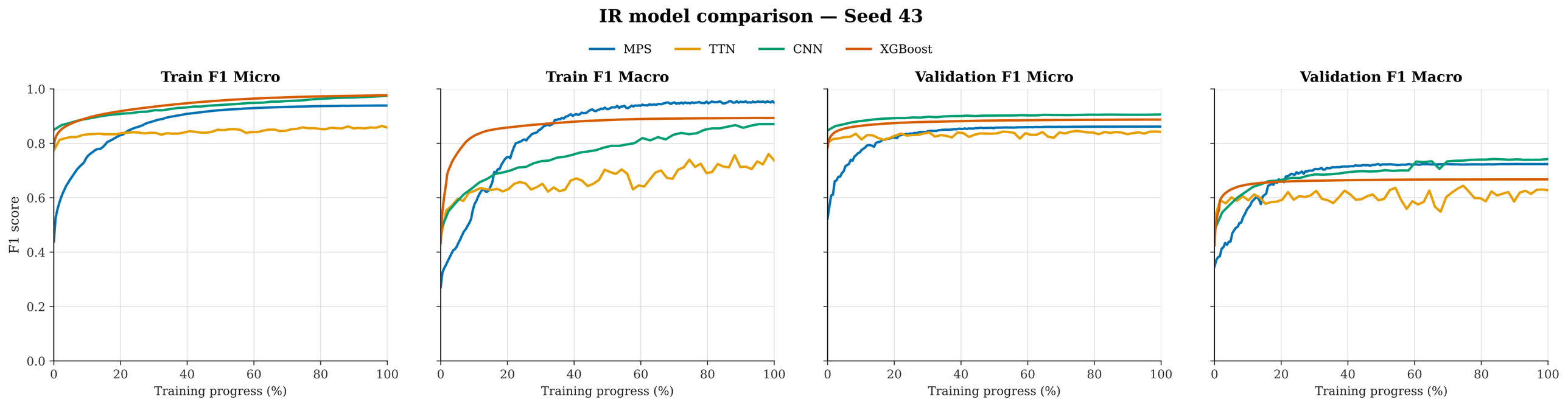

IR model comparison — Seed 42
MPS
TTN
CNN
XGBoost
Train F1 Micro
Train F1 Macro
Validation F1 Micro
Validation F1 Macro
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Training progress (%)
FIR + Gaussian smoothing + sqrt transformation + SNV normalization | Input dimension: 600

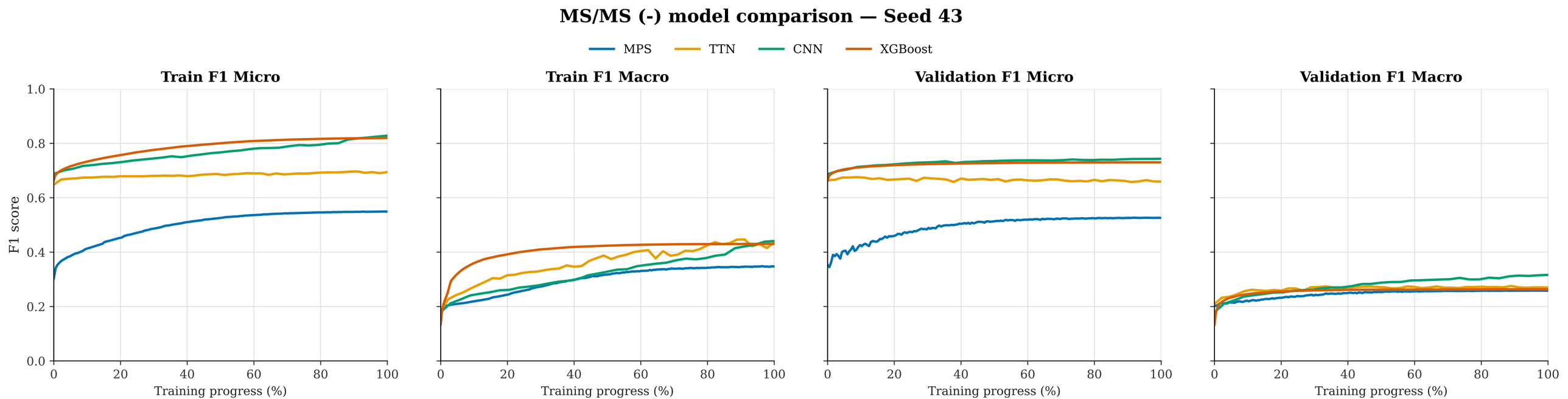

MS/MS (-) model comparison — Seed 42
MPS
TTN
CNN
XGBoost
Train F1 Micro
Train F1 Macro
Validation F1 Micro
Validation F1 Macro
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Training progress (%)
FIR + Gaussian smoothing + sqrt transformation + SNV normalization | Input dimension: 600

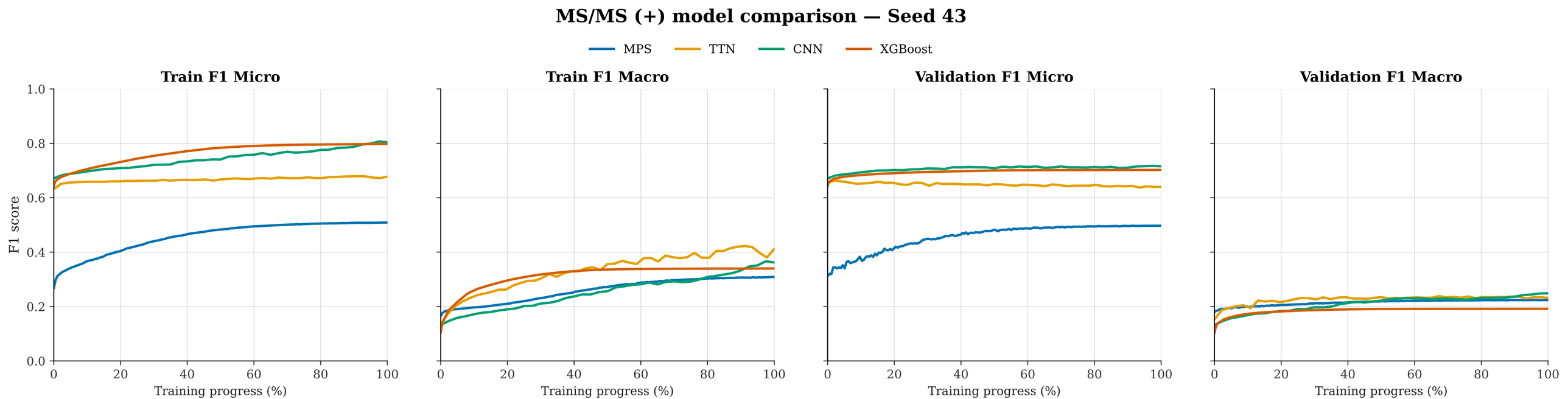

MS/MS (+) model comparison — Seed 42
MPS
TTN
CNN
XGBoost
Train F1 Micro
Train F1 Macro
Validation F1 Micro
Validation F1 Macro
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Training progress (%)
FIR + Gaussian smoothing + sqrt transformation + SNV normalization | Input dimension: 600


13C-NMR model comparison — Seed 43
MPS
TTN
CNN
XGBoost
Train F1 Micro
Train F1 Macro
Validation F1 Micro
Validation F1 Macro
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Training progress (%)
FIR + Gaussian smoothing + sqrt transformation + SNV normalization | Input dimension: 600


1H-NMR model comparison — Seed 43
MPS
TTN
CNN
XGBoost
Train F1 Micro
Train F1 Macro
Validation F1 Micro
Validation F1 Macro
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Training progress (%)
FIR + Gaussian smoothing + sqrt transformation + SNV normalization | Input dimension: 600


IR model comparison — Seed 43
MPS
TTN
CNN
XGBoost
Train F1 Micro
Train F1 Macro
Validation F1 Micro
Validation F1 Macro
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Training progress (%)
FIR + Gaussian smoothing + sqrt transformation + SNV normalization | Input dimension: 600


MS/MS (-) model comparison — Seed 43
MPS
TTN
CNN
XGBoost
Train F1 Micro
Train F1 Macro
Validation F1 Micro
Validation F1 Macro
F1 score
1.0
0.8
0.6
0.4
0.2
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20
40
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Training progress (%)
FIR + Gaussian smoothing + sqrt transformation + SNV normalization | Input dimension: 600


MS/MS (+) model comparison — Seed 43
MPS
TTN
CNN
XGBoost
Train F1 Micro
Train F1 Macro
Validation F1 Micro
Validation F1 Macro
F1 score
1.0
0.8
0.6
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40
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Training progress (%)
FIR + Gaussian smoothing + sqrt transformation + SNV normalization | Input dimension: 600